

Dr. Babasaheb Ambedkar Technological University, Lonere

Regular and Supplementary Examination Summer 2024

Course: B. Tech

Semester: VI

Branch: Electrical Engineering / Electrical & Power / Electrical & Electronics / Electronics and Power

Subject Code and Name: BTEEC603 Control Systems Engineering

Duration: 03 Hours

Date: 19/06/2024

Maximum Marks : 60

INSTRUCTIONS TO CANDIDATES

1. All questions are compulsory.
2. Assume suitable data wherever necessary and state the assumptions made.
3. Diagrams / sketches should be given wherever necessary.
4. Use of logarithmic table, drawing instruments and non-programmable calculators is permitted.
5. Use graph, semi-log paper to draw plots wherever required.
6. Figures to the right indicate full marks.

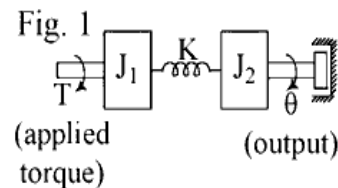
Q. 01 Attempt any two of the following

[CO,BL] (Marks)

- (a) Compare *open-loop* control system with *closed-loop* control system. [1,1] (06)
- (b) What is the effect of feedback on the following : (i) system gain (ii) system stability (iii) system sensitivity and (iv) noise or disturbance. Describe each in 1-2 sentence. [1,2] (06)
- (c) Find inverse Laplace for the following functions : [1,2] (06)
- (i) $F(s) = \frac{1}{s(s+1)}$ (ii) $F(s) = \frac{1}{s^2+4s+8}$

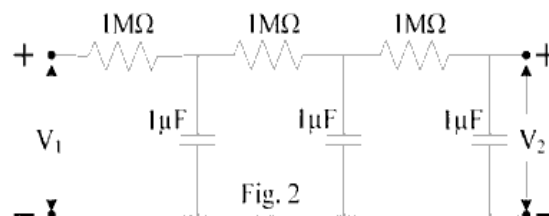
Q. 02 Attempt any two the following.

- (a) Obtain Transfer function for the mechanical-rotational system shown in fig. 1



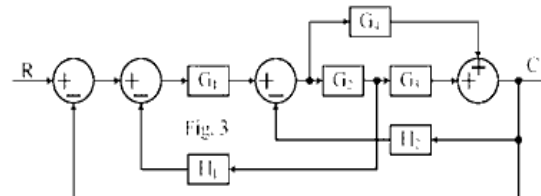
[2,3] (06)

- (b) Obtain Transfer function for the electrical network shown in fig. 1



[2,3] (06)

- (c) Using block-diagram reduction technique, obtain transfer function for the system shown in fig. 3



[2,3] (06)

Q. 03 Attempt the following.

- (a) Using time-response equation of standard second-order system, derive the equation for *Peak-Overshoot* (M_p) of the system. [2,3] (06)
- (b) The closed-loop transfer function of a certain system has *forward*, and *feedback* transfer function given by $G(s) = \frac{K_1}{s^2}$ and $H(s) = (1 + sK_2)$. Find the values of K_1 and K_2 so that the *peak-overshoot* of unit-step response is 25% and *peak-time* is 2 seconds. [2,1] (06)

Q. 04 Attempt any two of the following.

- (a) Applying Routh stability criterion, comment on the stability of the system whose characteristics equation is given by : $S^6 + 2S^5 + 8S^4 + 12S^3 + 20S^2 + 16S + 16 = 0$ [4,4] (06)
- (b) Sketch root-locus for the system whose open-loop transfer function is given by : $G(s)H(s) = \frac{K}{s(s+2)(s+3)}$. [4,4] (06)
- (c) Explain the effect of *phase-lag network* on the system performance with necessary equations and graphs. [4,2] (06)

Q. 05 Attempt any two of the following.

- (a) With necessary equations, and graphs explain the effect of *Proportional Controller* on system performance. [4,2] (06)
- (b) State the advantages of state-variable analysis. [4,1] (06)
- (c) Differential equation of a certain system is given by : $\ddot{y} + 6\dot{y} + 12y = 15\dot{u} + 25u$. [4,3] (06)
Obtain state-variable representation of the system.
